

Could dietary seaweed reverse the metabolic syndrome?



Incidence of the metabolic syndrome is increasing worldwide, with notable exceptions of some Asian countries where seaweeds are commonly consumed. 13 men (mean age 47.4 $\pm$ 9.9 yr) and 14 women (average age 45.6 $\pm$ 12.2 yr) with at least one symptom of the metabolic syndrome were recruited in Quito Ecuador to a randomized double-blinded placebo-controlled trial. Subjects were assigned to either Group 1 (1 m placebo, followed by 1 m 4 g/d seaweed [Undaria pinnatifida]) or Group 2 (1 m of 4 g/d seaweed, followed by 1 m of 6 g/d of seaweed). Blood pressure, weight, waist circumference, inflammation biomarkers, and lipids were measured monthly. Repeated measures analysis of variance with Tukey's multiple comparison tests were used for statistical analysis. In Group 2, systolic blood pressure decreased 10.5 mmHg after a month of 6 g/d seaweed (95% CI: 4.1, 16.8 mmHg; $p < 0.05$), primarily in subjects with high-normal baseline blood pressure. Waist circumference changed only for women participants, with a 2.4 cm decrease in Group 1 after treatment with placebo (95% CI: 1.0, 3.7 cm; $p < 0.01$). In Group 2, women had a mean decrease of 2.1 cm after 4 g/d (95% CI: 0.4, 3.7 cm; $p < 0.05$) and a further 1.8 cm decrease after 1 m 6 g/d seaweed (95 % CI: 0.1, 3.4, $p < 0.05$). No other changes were observed. Consumption of 4 to 6 g/d seaweed, typical for most people in Japan, may be associated with low metabolic syndrome prevalence.